NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR (Computer Science)

SPRING SEMESTER EXAMINATIONS 2024

Time : 3 Hours

Batch 2022

Dated : 25-JUL-24

Max Marks : 60

Dr. Umer Farooq

Database Management Systems - CT-261

Note: Attempt all questions. Answer all parts of a question in a row. State your assumptions clearly.

Q1. Normalize the attributes shown in the following table and *create* a set of well-designed relations up to third normal form (3NF). **[CLO 2: 15 Marks]**

Un-normalized Form:

Project Management Report

Project Code: PC010
Project Title: Pensions System

Project Manager: M Phillips **Project Budget:** £24,500

Employee No.	Employee Name	Department No.	Department Name	Hourly Rate
S10001	A Smith	L004	IT	£22.00
S10030	L Jones	L023	Pensions	£18.50
S21010	P Lewis	L004	IT	£21.00
S00232	R Smith	L003	Programming	£26.00

Total Staff on Project: 4 **Average Hourly Rate:** £21.88

Calculated Fields

Q2. a) *Explain* the states of data after commit and rollback with SQL commands. **[CLO 1: 5 Marks]**

b) *Generate* examples for the types of problem that can occur in a multi-user environment when concurrent access to the database is allowed. How can we prevent these problems with the help of Two Phase Locking? **[CLO 2: 10 Marks]**

Q3. a) *Discuss* the various types of NoSQL data models in detail with suitable examples. **[CLO 1: 5 Marks]**

b) What are the main types of threats that could affect a database system? *Describe* the controls and measures you would implement to counteract these threats and ensure database security. **[CLO 1: 10 Marks]**

Q4. a) *Explain* the concept of indexing in Database Management Systems. Illustrate the different types of indexing techniques with examples. **[CLO 1: 5 Marks]**

b) *Describe* the CAP Theorem and its significance in understanding the challenges of maintaining ACID-like guarantees in distributed systems. How does the CAP Theorem help in assessing the trade-offs involved in designing distributed databases? **[CLO 1: 10 Marks]**

CamScanner

SEAT NO. CT-21064

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
SECOND YEAR (BS COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
SPRING SEMESTER EXAMINATIONS 2023
Batch 2021

Dated : 05-AUG-23

Time : 3 Hours

Max Marks : 60

Database Management Systems - CT-261

Instruction:

i) Preferably, write the answers in sequential order.

ii) Illustrate your answers with neat sketches wherever necessary.

[CLO-1, 12marks]

Q.1 (a) List down any four types of joins.

(b) What type of locks are needed for insert and delete operations?

(c) What is a view? How it is related to data independence?

(d) What is cardinality and connectivity?

(e) Compare SQL and NoSQL database system.

(f) Compare DML, DDL and DCL statements.

(g) Compare Structured Semi-structured and Unstructured data.

Q.2 (a) Study the following detail and normalize the data up to 3rd normal form [CLO-2, 8 marks]

Project Management Report

Project Code: PC010
Project Title: Pensions System

Project Manager: M Phillips — Project Budget: £24,500

Employee No.	Employee Name	Department No.	Department Name	Hourly Rate
S10001	A Smith	L004	IT	£22.00
S10050	L Jones	L023	Pensions	£18.50
S21010	P Lewis	L004	IT	£21.00
S00232	R Smith	L003	Programming	£26.00

Total Staff on Project: 4 — Average Hourly Rate: £21.88

Calculated Fields

(b) Discuss insertion, deletion and modification anomalies in each normalized form of part a. [CLO-1, 4 marks]

Q.3 (a) A university have many faculties. The faculties in turn are divided into several departments. Each departments offers numerous programs and each program contains many courses. Lecturers can teach many different courses and even the same course numerous times. Courses can also be taught by many lecturers. A student is enrolled in only one program but a program can contain many students. Students can be enrolled in many courses at the same time and the courses have many students enrolled. Draw an ER diagram that captures this information with all possible attributes. [CLO-2, 8 marks]

(b) Discuss different type of indexes in Database management system also create index query. [CLO-1, 4 marks]

Q.4. (a) Solve any two-concurrency control problem using locking, with suitable examples. [CLO-2, 8 marks].

(b) Create a user with your name and assign system and object level privileges using SQL command. [CLO-2, 4 marks]

Q.5(a) Explain CAP Theorem and its practical guide for NoSQL database systems. [CLO-1, 8 marks]

(b) Differentiate between master slave and sharding in NoSQL database. [CLO-1, 4 marks]

SEAT NO. _____

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR(BACHELOR OF SCIENCE IN COMPUTER SCIENCE & INFORMATION TECHNOLOGY)

SPRING SEMESTER EXAMINATIONS 2022

BATCH 2020

Time: 3 Hours

Dated:17-08-2022

Max.Marks:60

Database Management Systems - CT-257

Instruction:

i) Preferably, write the answers in sequential order.

ii) Illustrate your answers with neat sketches wherever necessary.

Q.1.(a) List down any four types of joins. (12)

(b) What type of locks are needed for insert and delete operations.

(c) What is a view? How it is related to data independence?

(d) What is cardinality and connectivity?

(e) Compare SQL and NoSQL database system.

(f) What is DML (Data Manipulation Language)?

(g) Compare between Structured and Unstructured data.

Q.2.(a) How Entity-relationship helps in designing data tables? Give examples (3)

(b) Consider the tables given below:

Employees(employee_id,first_name,last_name,job_id,salary,department_d)

Departments(department_id,department_name,location_id) (9)

i) Find all employees who locate in the location with the id 1700.

ii) Finds all employees who salaries are greater than the average salary of all employees.

iiWrite a query to display the number of people with the same job.

Q.3.(a)Explain CAP Theorem and its implications for NoSQL database systems.(6)

(b) Discuss concurrency control problem and give example.(6)

Q.4 (a) What is the two-phase locking protocol? How does it guarantee serializability? What is transaction commit point? (6)

(b)Explain different type of indexes in Database management system also create index query. (6)

P.T.O

Q.5.(a) Study the following detail and normalize the data up to 3rd normal form (8)

Course Code	23	Department	Business
Course Title	Business Accounting		
Level	3	Tutor	Trevor Albright
Mode of Attendance	Part Time		
Course Fee	£55.00	Dept Head	Mr Alan Parkes
Start Date	20/08/2016		
End Date	24/08/2016		

Student ID Num.	Student Name	Student Address	Student Telephone	Student DOB	Enrolment Date	Enrolled By:-
14111	Alan Briggs	24 Abbey Park, Bangor, BT19 1RT	028 91468665 07798754623	10/08/94	10/06/16	Andrea Gilchrist
14332	Susan McBride	19 Belmont Road, Bangor, BT19 3TR	028 91488930 07891454235	01/03/93	03/07/16	David Smythe
14378	David Adair	14 Grange Park, Bangor, BT19 4TF	028 91584630 07891834277	06/11/87	05/07/16	Andrea Gilchrist

(b) Explain insertion, deletion and modification anomalies with suitable examples. (4)

SEAT NO. CT-19056

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR(BACHELOR OF SCIENCE IN COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
SPRING SEMESTER EXAMINATIONS 2021
BATCH 2019

Time: 3 Hours | **Dated:26-08-2021**
Max.Marks:60

Database Management Systems - CT-257

Instruction:
. Attempt all Questions.

Question NO 1:-

(a)What is fact-finding? (12)
(b)What type of locks are needed for insert and delete operations.
(c)What is a view? How it is related to data independence
(d) What is foreign key also write its query.
(e) The SQL statements used for transaction control.
(f) What is cardinality and connectivity?

Question NO 2:-

(a)Compare and contrast the two-tier client-server architecture for traditional DBMSs with the three-tier client-server architecture. Why is the latter architecture more appropriate for the Web? (6)

(b) The following tables form part of a database held in a relational DBMS (6)
Hotel (hotelNo, hotelName, city)
Room (roomNo, hotelNo, type, price)
Booking (hotelNo, guestNo, dateFrom, dateTo, roomNo)
Guest (guestNo, guestName, guestAddress)
where Hotel contains hotel details and hotelNo is the primary key;
Room contains room details for each hotel and (roomNo, hotelNo) forms the primary key;
Booking contains details of bookings and (hotelNo, guestNo, dateFrom) forms the primary key;
Guest contains guest details and guestNo is the primary key.
(a)Identify the foreign keys in this schema. Explain how the entity and referential integrity rules apply to these relations.
(b) Produce some sample tables for these relations that observe the relational integrity rules. Suggest some generalconstraints that would be appropriate for this schema.

Question NO 3:-

(a) Discuss concurrency control problem, and give examples (6)
(b) Explain a sparse index. Discuss the problem of dens index and sparse index. (3)
(c) Discuss the roles of the following personnel in the database environment (3)
i) Physical database designer ii) Application developer iii) End-users

P.T.O

Question NO 5:-

Study the following Project Management Report and normalize the data up to 3rd normal form, after normalization creates a query that joins all the tables. (12)

Project Management Report

Project Code: PC010
Project Title: Pensions System

Project Manager: M Phillips **Project Budget:** £24,500

Employee No.	Employee Name	Department No.	Department Name	Hourly Rate
S10001	A Smith	L004	IT	£22.00
S10030	L Jones	L023	Pensions	£18.50
S21010	P Lewis	L004	IT	£21.00
S00232	R Smith	L003	Programming	£26.00

Total Staff on Project: 4 **Average Hourly Rate:** £21.88

Calculated Fields

Question NO 4:-

(a) How Entity-relationship helps in designing data tables? Give examples (3)

(b) A database is to be designed for a Car Rental Co. (CRC). The information required includes a description of cars, subcontractors (i.e. garages), company expenditures, company revenues and customers. Cars are to be described by such data as make, model, year of production, engine size, fuel type, number of passengers, registration number, purchase price, purchase date, rent price and insurance details. It is the company policy not to keep any car for a period exceeding one year. All major repairs and maintenance are done by subcontractors (i.e. franchised garages), with whom CRC has long-term agreements. Therefore the data about garages to be kept in the database includes garage names, addresses, range of services and the like. Some garages require payments immediately after a repair has been made; with others, CRC has made arrangements for credit facilities. Company expenditures are to be registered for all outgoings connected with purchases, repairs, maintenance, insurance etc. Similarly the cash inflow coming from all sources - car hire, car sales, insurance claims - must be kept on file.CRC maintains a reasonably stable client base. For this privileged category of customers special credit card facilities are provided. These customers may also book in advance a particular car. These reservations can be made for any period of up to one month. Casual customers must pay a deposit for an estimated time of rental unless they wish to pay by credit card. All major credit card care accepted. Personal details (such as name, address, telephone number, driving licence, number) about each customer are kept in the database. (6)

(c).Discuss basic concepts of referential integrity .Give suitable examples (3)